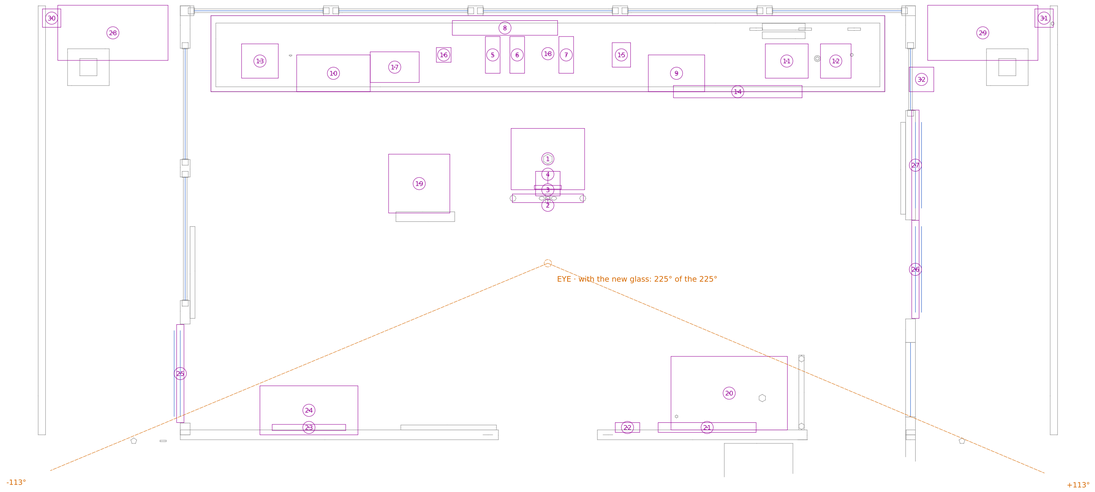
## PLAN · what she has (grey) and what it becomes (magenta), numbered as the table




## WHAT THE RANGER III PILOT HOUSE SHOWS (HAER MI-432 sheet 4, read at 75 px/ft, ±0.2 m)

· Her pilot house is 5.9 m wide and 4.8 m deep: the Christina's is 6.0 m by 3.5 m, the same width; a rounded glazed front buys the depth.
· Chart table 1.07 × 1.54 m, aft against the bulkhead (Christina: 0.95 × 0.60): kept, a half-fold chart fits.
· Steering stand 0.63 × 0.95 m on the centreline with a seat beside it (Christina: 0.50 × 0.95): widened.
· The master engine controls are ahead of the stand to port, the radar forward to starboard, the compass ahead of the stand: here the engine and thruster levers sit beside the stand, the radar to starboard, the compass overhead.
· A door to each wing, three steps up from the wing: the wing doors here are real, hinged, with a sill.
· The master's sea cabin is directly aft with its own stair; the Christina's master's cabin is on the corridor 3 m aft, a door apart: nothing to change.
· Rake: the forward glass leans out 10° at the top (head at yard 23.17; the compass deck's slab edge is 23.16, so lengthen it 0.01–0.10 m), against glare from the console at night. A guide value, recalled, not checked.
· View: with the two wing doors glazed in their upper halves and the starboard wall panel glazed, the arcs from the helm join into -113° to +113°; 225° of the 225° the navigation-bridge guide asks for.
· Nothing at eye level in the 20° right ahead: the heading readout is low in the stand, the compass is read overhead; the console's displays top at 1.33 m, which hides the sea only out to 50 m from the eye in a 20° arc to either side.
· Every item has its own supply (CHR-B4): the 24 V radio/emergency distribution feeds the VHF, telephone, alarm and lamps; a wheelhouse panel carries a fused way for each instrument.

## ITEMS · register id · item · position (x to port, yard, height above the deck) · what changes

**1 B073 Steering stand with wheel, NFU lever and autop**
x +0.00 yd 21.75 h 0.00 · widen to 0.60 (Ranger III 0.63); add the two missing controls

**2 B074 Helm wheel, eight spokes**
x +0.00 yd 21.43 h 0.76 · keep

**3 B075 Heading readout low on the stand's face**
x +0.00 yd 21.52 h 0.85 · MOVE from 1.50–1.62 m dead ahead to 0.85 m

**4 B084 Compass periscope eyepiece, overhead**
x +0.00 yd 21.55 h 2.05 · MOVE overhead (2.0–2.2 m); clears the horizon

**5 B076 Engine lever, port**
x +0.45 yd 22.60 h 1.00 · MOVE to x +0.45 (it was at +0.20: crossed)

**6 B077 Engine lever, starboard**
x +0.25 yd 22.60 h 1.00 · MOVE to x +0.25 (it was at +0.40)

**7 B078 Thruster lever**
x -0.15 yd 22.60 h 1.00 · ADD

**8 B083 Engine and heading repeater panel**
x +0.35 yd 22.82 h 1.15 · keep, tilt

**9 B081 Radar display**
x -1.05 yd 22.45 h 1.05 · LOWER the hood from 1.52 to 1.33 m; move 0.1 m inboard

**10 B082 Echo sounder and GPS plotter**
x +1.75 yd 22.45 h 1.05 · MOVE inboard, add the plotter on the same bracket

**11 B093 VHF with DSC and handset**
x -1.95 yd 22.55 h 1.00 · ADD

**12 B094 Ship's telephone and sound-powered phone**
x -2.35 yd 22.55 h 1.00 · ADD

**13 B095 NAVTEX printer**
x +2.35 yd 22.55 h 1.00 · ADD

**14 B085 Navigation-light panel (four double-pole switc**
x -1.55 yd 22.30 h 0.75 · MOVE from x −2.05 to −1.55 (VHF takes the end)

**15 B100 Searchlight handle**
x -0.60 yd 22.60 h 1.00 · ADD

**16 B099 Whistle push and fog-signal switch**
x +0.85 yd 22.60 h 1.00 · ADD

**17 B098 Ramp lamp board, lane signals**
x +1.25 yd 22.50 h 1.00 · ADD

**18 B089 Console, 5.5 m, top at 1.0 m**
x +0.00 yd 22.61 h 0.00 · keep

**19 B089 Master's high chair**
x +1.05 yd 21.55 h 0.00 · keep

**20 B087 Chart table, 0.95 × 0.60**
x -1.48 yd 19.84 h 0.00 · keep; lamp shade red

**21 B090 Fire and alarm panel**
x -1.30 yd 19.56 h 1.55 · keep; name it, give it a 24 V supply

**22 B097 PA and general alarm handset**
x -0.65 yd 19.56 h 1.35 · ADD

**23 B101 Clock, barometer, inclinometer**
x +1.95 yd 19.56 h 1.90 · ADD

**24 B102 Log desk (fold-down)**
x +1.95 yd 19.70 h 0.90 · ADD

**25 B104 Wing doors, hinged, glazed upper half, 0.15 m**
x +3.00 yd 20.00 h 0.00 · REPLACE the plates laid beside the doorways

**26 B104 Starboard wing door, same**
x -3.00 yd 20.85 h 0.00 · REPLACE

**27 B105 Starboard wall panel glazed, 0.90 m**
x -3.00 yd 21.70 h 0.97 · ADD glass

**28 B107 Port wing console, 0.90 × 0.45**
x +3.55 yd 22.78 h 0.00 · ADD

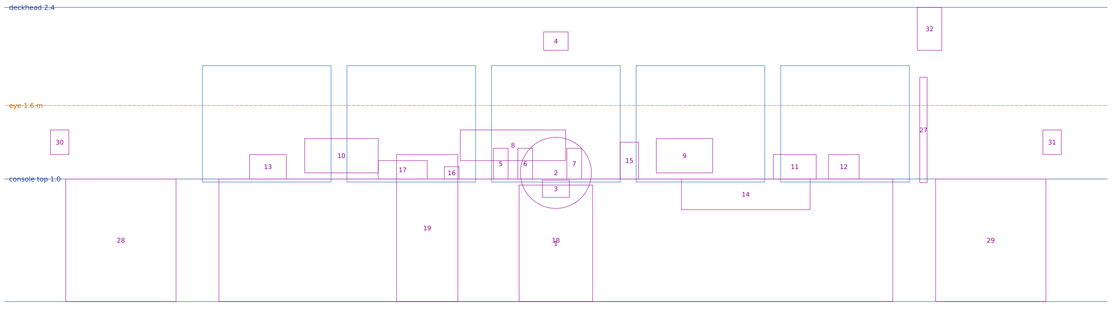
**29 B109 Starboard wing console**
x -3.55 yd 22.78 h 0.00 · ADD

**30 B108 Port sidelight (red) on a screen**
x +4.05 yd 22.90 h 1.20 · MOVE to the wing from the exterior

**31 B110 Starboard sidelight (green) on a screen**
x -4.05 yd 22.90 h 1.20 · MOVE

**32 B111 Float-free EPIRB in its bracket**
x -3.05 yd 22.40 h 2.05 · ADD

## FROM THE WHEEL, LOOKING FORWARD · the same items by height




Nothing stands at the eye's level in the 20° right ahead; the highest thing on the console is the radar, at 1.33 m.

Drawn from ferry/data/ferry.bin.gz and the Town data at the commit below; not for construction
**THE CHRISTINA · bridge survey, Rev A**

# A REALISTIC WHEELHOUSE · design basis

Every item of the design words placed, the view made whole; the envelope is the

Scale 1:25

Metres on the sheet at the stated scale; DXF in sheet millimetres
ferry.bin.gz · Town f0b2973

Drawing no.
**CHR-B6**